CHAPTER

Environmental and safety issues of lead–acid batteries

9

Introduction

During the whole life cycle of the lead–acid battery (LAB) starting with materials via production, use, and finally end of life, the battery has to fulfill all environmental and safety requirements.

The environmental challenges are mainly related to lead and lead compounds, sulfuric acid spilling, toxic gases developed during operation, and the CO_2 footprint during production, use, and end of life. Safety issues are related to thermal, chemical, kinetic, and electrical hazards (Fig. 1).

Additional to these hazards is to consider the loss of functionality of a battery driven equipment in case of failure of the battery.

Because the chemical hazards are closely linked to the environmental challenges, they will be discussed in the environmental section.

Environmental challenges

Environmental problems are divided into human health and ecological problems.

Health risks are related to adverse health effects after humans are exposed to environmental hazards, e.g., via inhalation, ingestion, or dermal contact, while ecological risks are focused on the risk borne by ecosystems and their components.

The environmental issues of lead–acid batteries are based on the battery components lead (grid), lead compounds (negative (Pb) and positive (PbO_2) active mass) and the electrolyte (sulfuric acid). The problems with lead and its compounds occur during mining, melting, battery production, recycling and disposal, while problems with the electrolyte are observed mainly during use and recycling as well as disposal.

Normally the lead–acid battery itself when used according to instructions does not create harmful substances as lead dust, vapors, or fumes. But this can be observed if the battery will be overheated or damaged (see the safety section).

The challenges are mainly related to the release of substances harmful to health and/or the environment. The hazard evaluation is done according to NFPA 704 (US-based Standard System for the Identification of the Hazards of Materials for Emergency Response) and to GHS (Globally Harmonized System of Classification and Labelling of Chemicals).

The NFPA 704 system evaluated hazards related to health (blue), flammability (red), reactivity (yellow), and special notice (white). The NFPA 704 numbers (0, 1, 2, 3, and 4) characterize the severity of potential problems with flammability, reactivity, and health, whereas number 4 has the highest severity. The "special notice" could be empty (no special attention), OX (oxidizer allows the chemical to burn without air supply), and W (reacts with water in an unusual and dangerous manner—see Figs. 2, 3–5, and 7–10.

The GHS classifications used in this chapter are shown in general in Table 1.

Advanced Industrial Lead–Acid Batteries. https://doi.org/10.1016/B978-0-444-63252-4.00009-9
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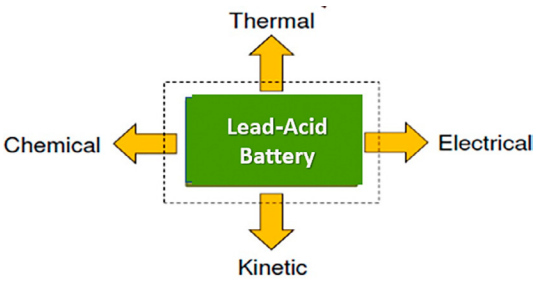


FIG. 1

Different hazards of lead–acid batteries.

Table 1 GHS classifications.

GHS 02		Are flammable; liquids form explosive mixtures with air; produce flammable gases with water or are self-ignitable
GHS 03		Are oxidizing and increasing fires, when mixed with combustible substances, explosive mixtures are formed
GHS 05		Are corrosive to metals, dermocaustic, and could show severe eye damage
GHS 06		Leading immediately to serious damage or death in smaller quantities
GHS 07		Lead to health damage and irritate eyes, skin, or respiratory organs. Leading to death in larger quantities

Table 1 GHS classifications—cont'd		
GHS 08		Cause allergy, carcinogenic and mutagenic, toxic for reproduction, and teratogenic (reprotoxic) or organ damaging
GHS 09		Shows environmental hazards, i.e., are harmful to aquatic organisms, toxic or very toxic, acute or with long-term effects

Issues related to lead and lead compounds

Lead

The hazard classification pictograms for lead related to NFPA 704 is shown in Fig. 2A and related to GHS in Fig. 2B.

Lead is responsible for many health problems [1]. Among others, the following health problems with Pb have to be considered:

- Intellectual deficits
- Aggressive behavior
- Mood disorders
- High blood pressure
- Joint and muscle pain (colic)
- Reduced sperm count and abnormal sperm

The blood lead level (BLL) of the lead worker was in average very high before the lead health danger was universally recognized and laws were introduced as limitations in the EU—EU-98/24, 1998—70 μg dL^{-1}; limitations in Germany—Technische Regeln für Gefahrstoffe TRGS 903, 2000—40 μg dL^{-1}; or limitations of the International Lead Association (ILA), 2017 – 20 μg dL^{-1}.

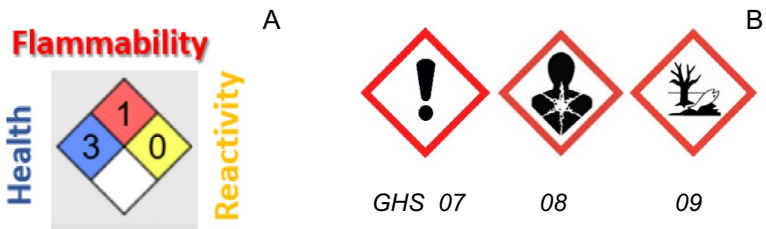


FIG. 2

Hazard classification pictograms for Pb related to NFPA 704 (A) and GHS 07, 08, 09 (B).

The large advances in reducing the BLL of lead workers goes back to improved process technology and health protection. Nevertheless, progress in BLL has not yet been achieved worldwide; in some Asian countries in particular, there are still backyard factories that operate without adequate health and safety standards.

Lead compounds

The most important lead compounds PbO_2 and $PbSO_4$ in the LAB technology are from health and environmental point of view arranged analogously to lead—see Figs. 2, 3, and 4.

Issues related to the sulfuric acid electrolyte

The hazard classification pictograms for H_2SO_4 related to NFPA[a] are shown in Fig. 5A and related to GHS[b] in Fig. 5B.

The following health problems with H_2SO_4 are to consider [2]:

Inhalation	Coughing, shortness of breath, difficulty breathing, and can cause death
Skin contact	Pain, redness, burns, and blistering
Eye contact	Pain, redness, swelling, blindness
Ingestion	Nausea, vomiting, stomach cramps, diarrhea, and can cause death



FIG. 3

Hazard classification pictograms for PbO_2 related to NFPA 704 (A) and GHS 03, 08, 07, 09 (B).

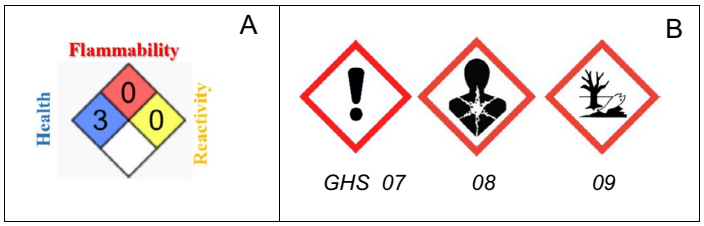


FIG. 4

Hazard classification pictograms for $PbSO_4$ related to NFPA 704 (A) and GHS 07, 08, 09 (B).

[a]US based Standard System for the Identification of the Hazards of Materials for Emergency Response.
[b]International Globally Harmonized System of Classification and Labeling of Chemicals.

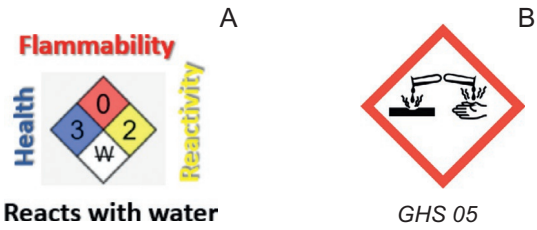


FIG. 5

Hazard classification pictograms for H_2SO_4 related to NFPA 704 (A) and GHS 05 (B).

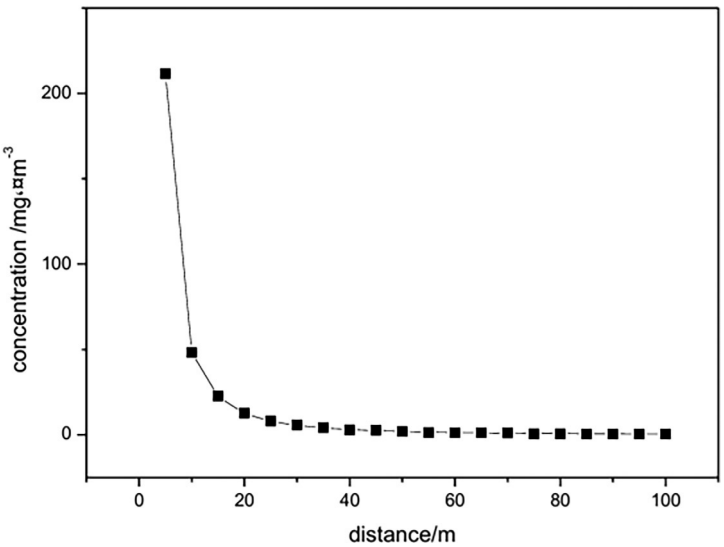


FIG. 6

After LAB explosion—H_2SO_4 concentration in air vs distance of explosion place [3].

The impact on the environment is not only caused directly by the electrolyte being spilled onto the ground, but also via the air in case of a LAB explodes, whereby the latter is also influenced by the wind speed. Fig. 6 shows a calculation of the spread of sulfuric acid after an explosion at a wind speed of 1 m/s.

Issues related to toxic gases

During the operation of the LABs under certain conditions, such as high temperatures, high current and high voltages, toxic gases are generated in addition to oxygen and hydrogen gases.

SO_2, H_2S

Sometimes the development of SO_2 and H_2S during charging of the negative electrode by reduction of the sulfuric acid is observed. For the formation of these toxic gases, the following mechanism could be possible, although the exact process is not completely cleared up.

SO_2 development:

$$\begin{aligned} &HSO_4^{2-} + 3H^+ + 2e^- \Rightarrow \mathbf{H_2SO_3} + H_2O \\ &\mathbf{H_2SO_3} \Rightarrow \mathbf{SO_2} + H_2O \end{aligned} \tag{1}$$

H_2S development:

$$\begin{aligned} &\mathbf{H_2SO_3} + 4H^+ + 4e^- \Rightarrow \mathbf{S^0} + 3H_2O \\ &\mathbf{S^0} + Pb^0 + 4H^+ \Rightarrow \mathbf{H_2S} + H_2 + Pb^{++} \\ &\mathbf{S^0} + 2H^+ + 2e^- \Rightarrow \mathbf{H_2S} \end{aligned} \tag{2}$$

There are also other reaction steps, e.g., could be the H_2SO_3 formation runs via the hypo disulfuric acid [4]. All intermediate reaction steps have a more positive potential as the negative Pb electrode, i.e., from the thermodynamic point of view, the formation of SO_2 and H_2S is possible. But under normal LAB battery operating conditions, no formation of these toxic gases is observed.

The hazard classification of these gases is shown in Figs. 7 and 8.

Both gases (SO_2, H_2S) are toxic, whereby the toxicity of H_2S is with the NFPA 704 classification "Health = 4" even larger as of SO_2 with "Health = 3."

There are only a few papers [5–9] in the literarture about SO_2 and H_2S development.

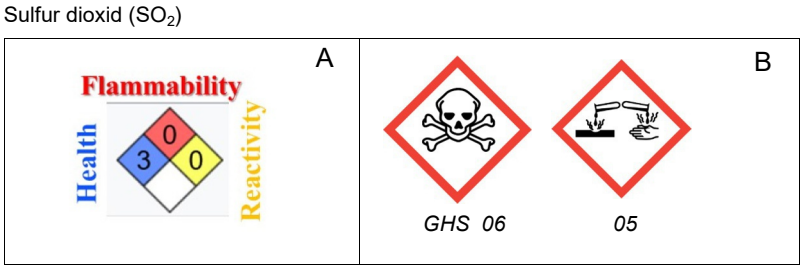


FIG. 7

Hazard classification pictograms for SO_2 related to NFPA 704 (A) and GHS 06, 05 (B).

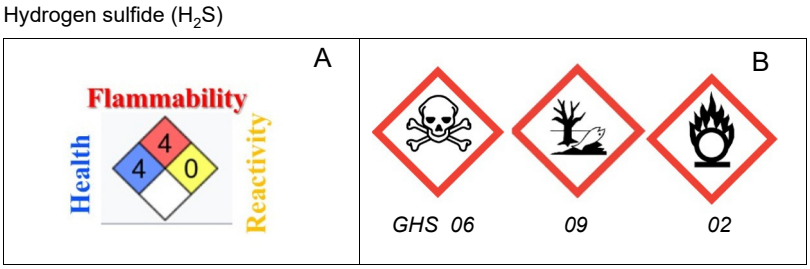


FIG. 8

Hazard classification pictograms for H_2S related to NFPA 704 (A) and GHS 06, 09, 02 (B).

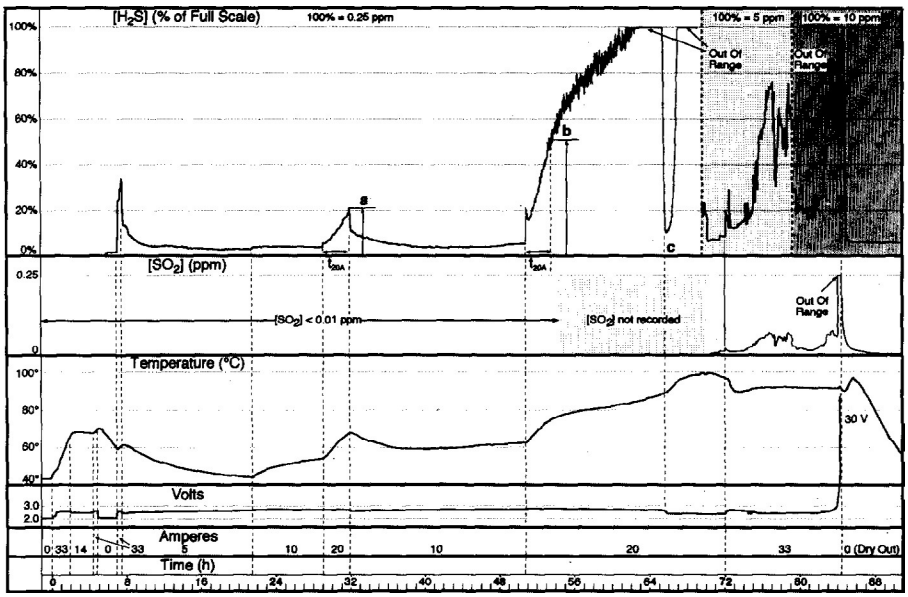


FIG. 9

Charge of a 160 Ah VRLA cell with different charge currents (*I*) and measurement of voltage (*U*), temperature (*T*), concentration of sulfur dioxide (SO_2) and hydrogen sulfide (H_2S); the *dashed lines* extending to the time axis indicate when the cell current was changed.

Modified from Robinson RS, Tarascon JM. J Power Sources 1994;48:277–284.

Fig. 9 shows the influence of the charge current of a VRLA cell on the SO_2 and H_2S concentration.

Fig. 9 shows that H_2S is developed at higher currents and the associated higher temperatures (>60°C), see 7 h and 33 A, 29 h and 20 A, and 51 h and 20 A. At lower currents of <10 A and lower temperatures <60°C, the H_2S development rate was reduced.

The temperature influence on the H_2S development rate is shown for the transition from a 10 to 20 A charge current at 29 and 51 h. At 51 h, the cell temperature is 65°C already 10K higher than as at 29 h and the H_2S development started instantaneously after the current increase.

After the current increase (10 A => 20 A) at 51 h the cell was then subjected to prolonged, extreme overcharge, which was further amplified by a charge current increase to 33 A. current. This led to an increase of the temperature and therefore higher H_2S development rates. At $t = 65.5$ h unfortunately the cell sealing valve failed (point c in Fig. 9) and led to a drop in the H_2S concentration probably by the interference of water vapor. The open valve leads finally to a cell drying out with a dramatic increase of the voltage and the H_2S concentration (the upper measurement level of 10 ppm was by far exceeded and values of 30…50 ppm were estimated based on the C_{H2S} peak shape). The olfactory threshold amounted to 0.13 ppm.

By the current increase to 33 A at about 72 h also SO_2 development was observed, the c_{SO2} reached the upper measurement level (0.25 ppm) but concentrations of a few ppm were appraised.

In the evaluation of the H_2S and SO_2 concentration, it is considered that the formed H_2S and SO_2 could during charge be oxidized at the positive electrode [8].

As mentioned previously, the reduction of H_2SO_4 to SO_2 and H_2S is possible from the thermodynamic point of view. But under normal operating conditions, these reactions are strongly hindered. As the experimental results have shown (Fig. 23) an increase in temperature, current, and potential leads to a reduction of the kinetic hindering. That also means that all conditions leading to a thermal runaway promote the development of these gases [10].

SbH_3, AsH_3

Stibine (SbH_3) and arsine (AsH_3) are forming in LAB batteries with PbSb and PbSbAs grids during charge.

SbH_3 development:

$$Sb + 3H^+ + 3e^- \Rightarrow \mathbf{SbH_3} \tag{3}$$

AsH_3 development:

$$As + 3H^+ + 3e^- \Rightarrow \mathbf{AsH_3} \tag{4}$$

The hazard classification of these gases is shown in Figs. 10 and 11.

In reactions Eqs. (3) and (4) at the negative electrode react mainly to the Sb and As large area deposited on the negative active mass from Sb^{3+} and As^{3+} in the electrolyte, which are formed as corrosion products of the positive antimonial grid.

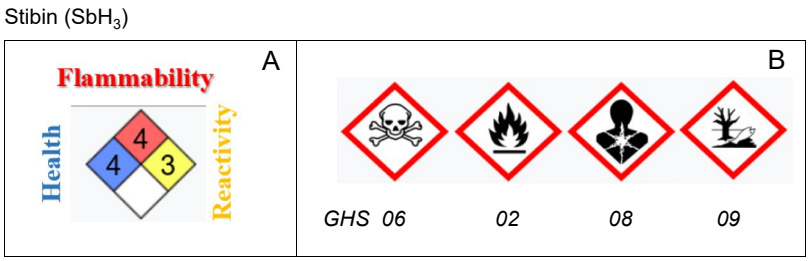


FIG. 10

Hazard classification pictograms for SbH_3 related to NFPA 704 (A) and GHS 06, 02, 08, 09 (B).

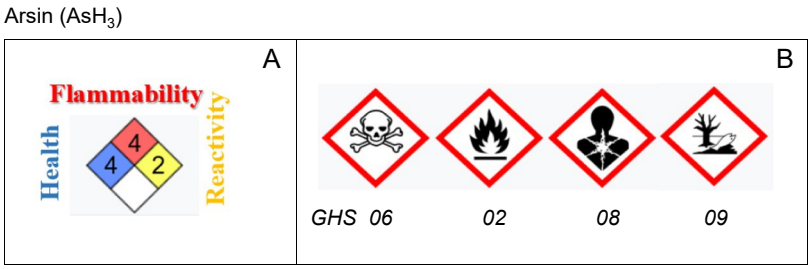


FIG. 11

Hazard classification pictograms for AsH_3 related to NFPA 704 (A) and GHS 06, 02, 08, 09 (B).

The generation of SbH_3 and AsH_3 started at cell voltage >2.45 V—see Fig. 12.

Because SbH_3 and AsH_3 are reducing the hydrogen overvoltage at the negative electrode, sometimes boost charges with ≥2.6 V are carried out to reduce the Sb and As content on the negative electrode by producing SbH_3 and AsH_3—for good ventilation to take care. This boost charge is to repeat from time to time because Sb and As will be further deposited caused by the corrosion of the positive grid. The deposition rate, however, is reduced with the operating time [11,12].

To avoid the formation of the toxic gases SbH_3 and AsH_3 the charge voltages should be <2.45 V. The problem can also be solved by using lead alloys without Sb and As, which is today the case in most SLI batteries.

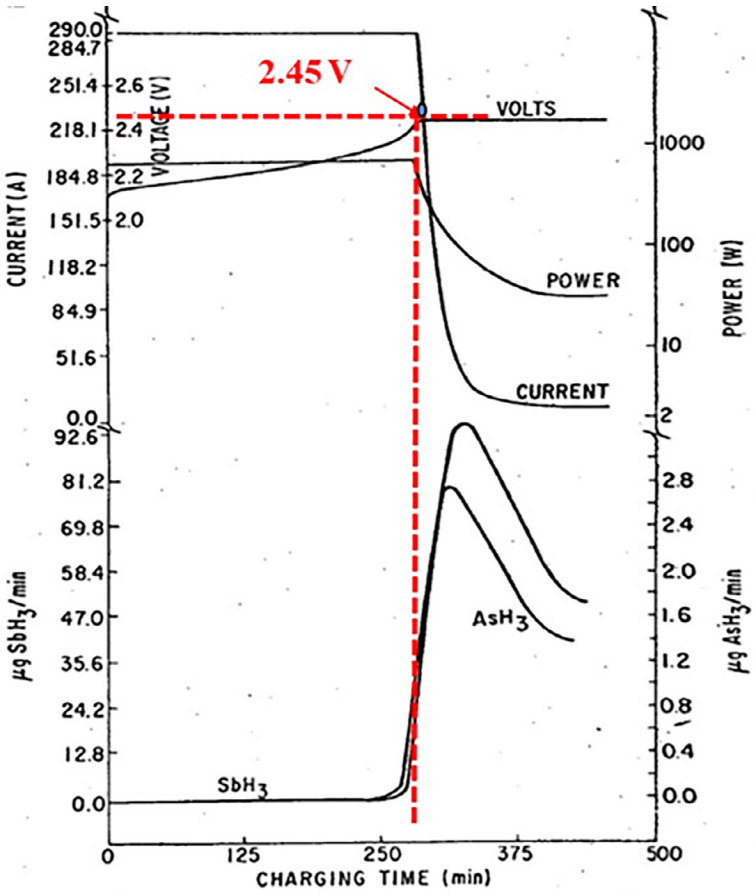


FIG. 12

IU charge of 2250 Ah LAB cells with positive grid PbSb4.6As0.08 and negative grid PbSb4.3As0.08 with development rates of SbH_3 and AsH_3.

Modified from Varma R, Cook GM, Yao NP, Argonne National Laboratory report, Argonne National Lab., IL; Department of Energy, Washington, DC, Technical Report, NTIS Issue, Number 198025, Contract Number W-31-109-ENG-38, 1978.

In general, for reducing the hazard of toxic gases (SO_2, H_2S, SbH_3, and AsH_3) if the operating conditions of the LAB are not considered a good ventilation of the battery rooms has to be guaranteed. A development of toxic gases at normal operating conditions is not to fear! Furthermore, it is considered that safety against toxic gases in stationary storage units is ensured based on strong ventilation requirements due to the formation of explosive gases (H_2, O_2).

Environmental regulations

In order to reduce the negative environmental effects general of batteries with their direct harmful emissions, but also to minimize the energy and the associated CO_2 emissions for production and disposal, a number of measures have been adopted.

First and foremost, the new EU battery regulation [13] should be mentioned here which entered into force in August 2023. This regulation applies to batteries of all chemistries and therefore also for LABs and all applications and therefore also for industrial use.

The key features of the battery regulation are:

- Due diligence obligations to address social and environmental risks
- Recycled content min. Requirements for industrial, EV, and automotive batteries
  From August 2031: 16% for Co, **85% for Pb**, 6% for Li, and 6% for Ni
  From August 2036: 26% for Co, **85% for Pb**, 12% for Li, and 15% for Ni.
- Digital passports with carbon footprint declarations and compliance with the maximum carbon footprint threshold (will be fixed until 2027)
- Stricter targets for waste collection, recycling efficiency, and material recovery
- All collected batteries will need to be recycled, and high levels of recovery must be achieved:
  Li: 50% by the end of 2027 and 80% by the end of 2031
  Co, Cu, Pb, and Ni: 90% by the end of 2027 and 95% by the end of 2031

In principle, all stages along the value chain up to recycling are included in the regulation, and so a circular economy of batteries is demonstrated—see Fig. 13.

In the past, most of the environmental contamination by lead, which was mainly related to lead–acid batteries was airborne. By strict regulations in the mining, melting, lead–acid battery production, and recycling, this part was successful reduced. However, water contamination by lead still remains a concerning issue and a monitored parameter of environmental health. A new EU drinking-water directive from 2021 gives a limit of max $5\,\mu g\,L^{-1}$, which must be met by 12 January 2036 at the latest. The parametric value for lead until that date is $10\,\mu g\,L^{-1}$ [14]. This is in line with legislation in several other countries. Problems in general with regulations and their critical values are the availability of analytical instruments with the appropriate detection limit. For example, is the common flame atomic absorption spectrometry (FAAS) with a detection limit of $\sim 10\,\mu g\,L^{-1}$ not sufficient for the EU goal of $5\,\mu g\,L^{-1}$. Other methods such as anodic stripping voltammetry (AVS), graphite furnace atomic absorption spectrometry (GFAAS), or inductively coupled plasma mass spectrometry (ICP-MS) are more accurate, but also require some laboratory expertise and are considerably more expensive [15]. In addition, it is to mention, that the analysis of nanoparticles is generally not yet sufficiently developed. These analytical problems particularly in low- and middle-income countries (LMICs), mean that environmental standards cannot be implemented in practice. An overview [16] dated 2016 estimates that there are in 90 countries ca.10,000–30,000 informal used lead–acid battery processing sites which influence the health

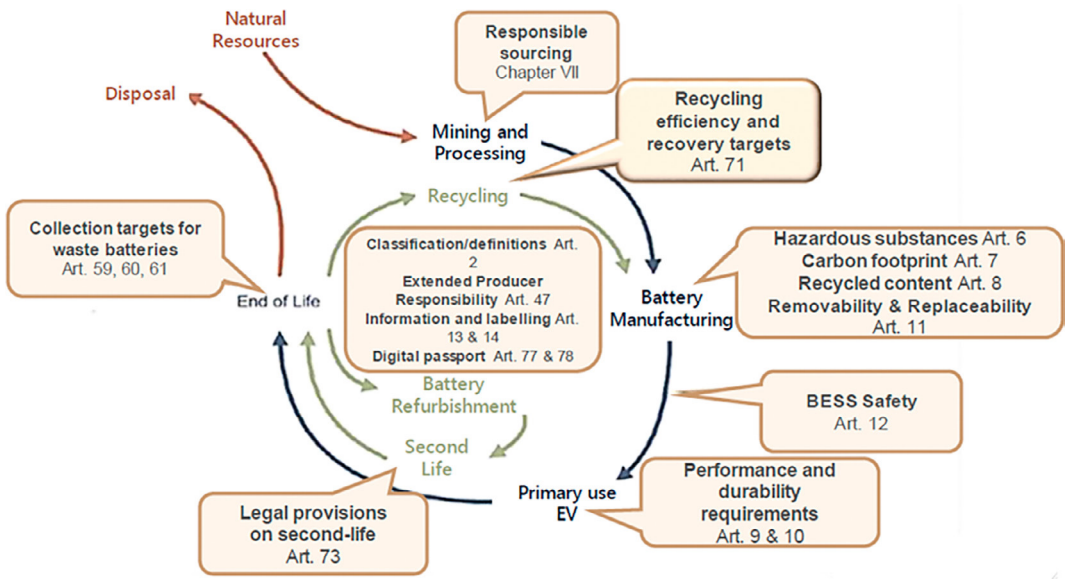


FIG. 13

Battery circular economy related to different articles of the new EU battery regulation.

Source EU Commission.

of 6–17 million people. The calculated blood lead levels for children (0–4 years of age) amounted to 31.15 mg dL^{-1} and for adults of 21.2 mg dL^{-1}.

Finally, it is in the lead–acid battery environmental context to mention that the EU has adopted the *European Directive 2000/53/EC on end-of-life vehicles* already in the year 2000. This directive regulates the handling of all strong hazardous to health components of vehicles at their end of life and may even prohibit the use of such components in vehicles. The directive allows, however, exceptions as long as no other competing products with higher health and environmental standards are available. Such an exception is also valid for SLI lead–acid batteries. The EU is proofing this exception every 5 years, the last review was done in 2023. The question is of course if and when the EU will accept Li-ion batteries as a more environmentally friendly competitor, which has, however, for the time being not yet a sufficient cold-cranking capability.

Recycling

An important issue in the lead–acid battery industry is the recovery of lead and the disposal of battery waste. Environmental issues have become increasingly important over the last three decades, and the toxicity of lead has brought the LAB into the spotlight. These objections have been partly counterbalanced by the need for a low-cost storage technology for renewable energies from wind and photovoltaics.

The widespread deployment of VRLA cells, particularly the AGM type, in the last 30 years, required the use of soft lead for lead dust and antimonial-free alloys with significantly high purity levels.

This task could only be achieved by selectively reducing the content of impurities in the leady oxide to very low values for elements such as Te, Se, Ni, As, and Cu, which catalyze the decomposition of water at high potential [17] and leads to dry-out mainly of VRLA-AGM cells due to H_2O losses in float-charge operation, particularly at elevated temperatures.

These very low levels of residual impurities posed an additional challenge, especially for secondary lead smelters, which led to an increase in metal processing costs. Table 2 provides a qualitative and quantitative overview of the type and proportion of lead battery materials involved in secondary lead refining [17].

There are two options for recycling LABs: the pyrometallurgical and the hydrometallurgical electrolytic process.

Pyrometallurgical process

A survey on the processing steps involved in pyrometallurgical recycling of secondary lead is shown in Fig. 14.

After the initial battery breaking and component separation, the following step is:

Desulfurization

For example, by heating the battery scrap:

$$PbSO_4 + C + \text{Heat} \rightarrow PbO + SO_2 + CO \tag{5}$$

Alternatively:

$$PbSO_4 + 4C \rightarrow PbS + 4CO \tag{6}$$

at $T > 600°C$

$$PbS + Na_2CO_3 \rightarrow PbO + Na_2S + CO_2 \tag{7}$$

The next step is the battery scrap smelting.

Table 2 Breakdown of materials in spent lead–acid batteries.

Component	Percentage by weight
Pb (grids and poles)	25
Active material	
$PbSO_4$	50–60
PbO_2	15–35
PbO	5–10
Metallic Pb	2–5
Other	2–5
Total active material	54
Polypropylene cases	5
Separator, etc.	5
H_2SO_4 (15%)	11

Reproduced from Ellis TW, Mirza AH. The refining of secondary lead for use in advanced lead-acid batteries. J Power Sources 2010;195:4525–4529.

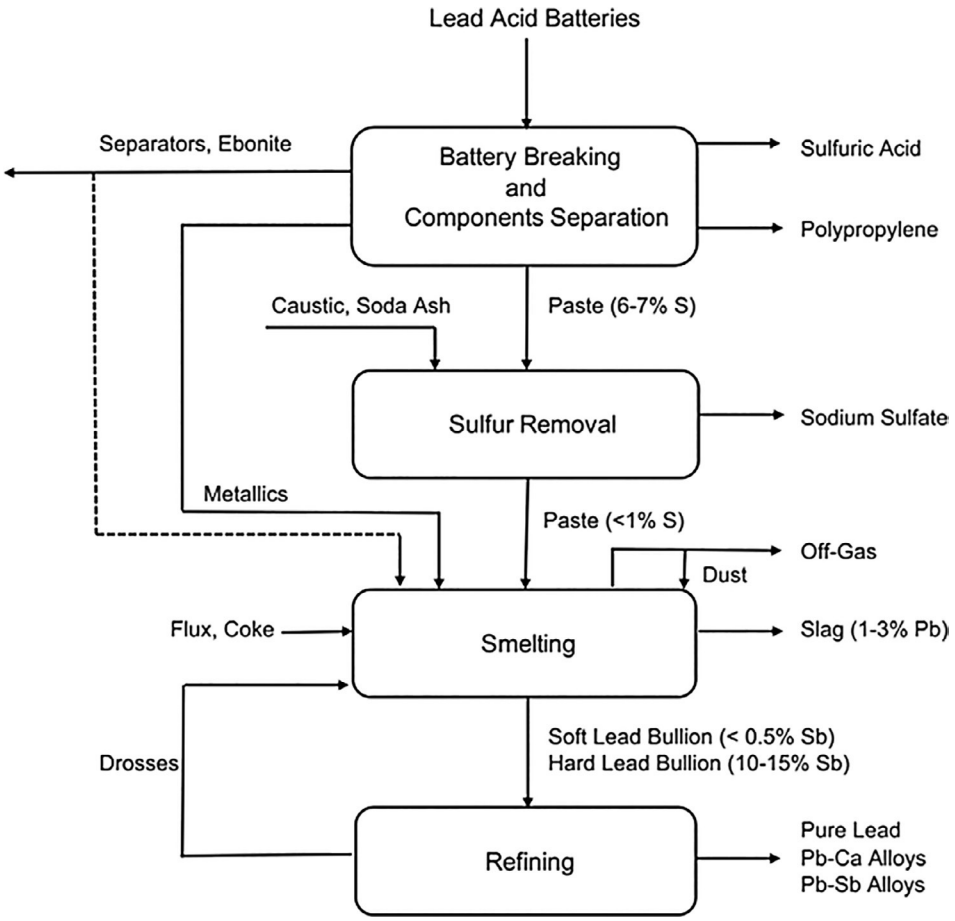


FIG. 14

Flow diagram for secondary lead recycling.

Reproduced from Ellis TW, Mirza AH. The refining of secondary lead for use in advanced lead-acid batteries. J Power Sources 2010;195:4525–4529.

Smelting

Lead recyclers around the world usually use rotary drum furnaces for secondary lead smelting. See Hoppecke's rotary furnace in Fig. 15. These furnaces are flexible and easy to operate, allowing the processing of the metal bullion in a single step [18].

The main reaction in the furnace is the oxide reduction with coke [19,20]:

$$PbO + C \rightarrow Pb + CO / CO_2 \tag{8}$$

Alternative neutralization with soda ash (Na_2CO_3)

$$PbSO_4 + Na_2CO_3 \rightarrow PbCO_3 + Na_2SO_4 \tag{9}$$



FIG. 15

Front view of a rotary drum furnace showing the closure (left) for the molten lead tapping (right).

in the furnace:

$$PbCO_3 + Heat \rightarrow PbO + CO_2 \quad (10)$$

and again

$$PbO + C \rightarrow Pb + CO / CO_2 \quad (8)$$

Lead refinement

After smelting the next step is the removal of residual elements from the bullion to produce soft lead and lead alloys. The degree of purity required depends on the different material specifications, i.e., leady oxide for the PAM and NAM, grids, and terminal alloy compositions.

Pyrometallurgical lead refining processes are specific to the different residual elements. Some of these (Harris, Parkes) were already developed in the nineteenth century or at the beginning of the last century (Kroll-Betterton).

The removal of residual elements from the lead bullion proceeds in different refining steps [19,20]:

- Cu is firstly removed by stirring sulfur into the pot at ca. 330°C or with iron pyrite (FeS_2)
- Ni removed with FeS_2 and S
- Sb oxidized by blasting air into the pot (softening)
- As and Sn are also oxidized by blasting oxygen-enriched air into the pot at 600°C
- Fe and Zn can also be removed by oxidation
- Sb is also oxidized with sodium nitrate ($NaNO_3$) or sodium hydroxide Na(OH)—Harris process
- Te is removed with KOH or NaOH

- Ag (and another noble metal) removed with Zn addition (Parkes desilverizing process)
- Bi remove with Ca and Mg addition (Kroll-Betterton process)

The removal of Ag and Bi is a standard process for primary lead smelters. In secondary lead recycling, however, the removal of Ag and Bi is not profitable due to the high costs involved for secondary smelters.

Environmental concerns

Environmental concerns against the pyrometallurgical recycling process are:

(i) Gas emissions: mainly sulfur dioxide. SO_2 is captured in off-gas scrubbing by reaction with carbonates or hydroxide in aqueous solution [18].

(ii) The disposal of the slag. The slag consists of the ash fraction of the coke, oxidized iron, and a minor amount of PbO, i.e., molten oxides and sulfides [21]. The general opinion is that extracting metals from slag is uneconomical, although there are people working on slag recovery as it is an important issue [22].

(iii) Health hazards for employees in the lead industry [22].

(iv) Lead smelting has a high energy demand due to the relatively high operating temperatures, while the use of carbon as a fuel leads to the generation of CO_2. The high energy demand in conjunction with the production of CO_2 means that lead smelting has a comparatively high global warming potential (GWP) [23].

To prevent environmental pollution from lead, the contaminated air and water must be extracted after lead removal using dust collectors and water treatment systems.

Hydrometallurgical process

Electrolytic hydrometallurgical processes can potentially produce lead of greater purity than conventional secondary pyrometallurgical methods without passing through a smelting and refinery process generating fewer waste products [22]. The widespread use of these methods was hampered by the high cost of electricity and consumable chemicals. The prospects, however, will also depend on the evolution of the increasingly tough environmental regulations.

The working principle for some of these electrowinning methods consists of dissolving the lead sludge in strong acids like HCl (PLACID process—see Fig. 16), HBF_4 (Engitec Flubor—primary lead), or methanesulfonic acid (Aqua Metals).

Reaction pathways for the PLACID electrowinning process

$$PbO + 2HCl \rightarrow PbCl_2 + H_2O \tag{11}$$

$$Pb + PbO_2 + 4HCl \rightarrow 2PbCl_2 + 2H_2O \tag{12}$$

$$PbSO_4 + 2NaCl \rightarrow 2PbCl_2 + Na_2SO_4 \tag{13}$$

Cathode:

$$PbCl_2 + 2e^- \rightarrow Pb + 2Cl^- \tag{14}$$

Anode:

$$H_2O \rightarrow 2H^+ + 1/2\,O_2 + 2e^- \tag{15}$$

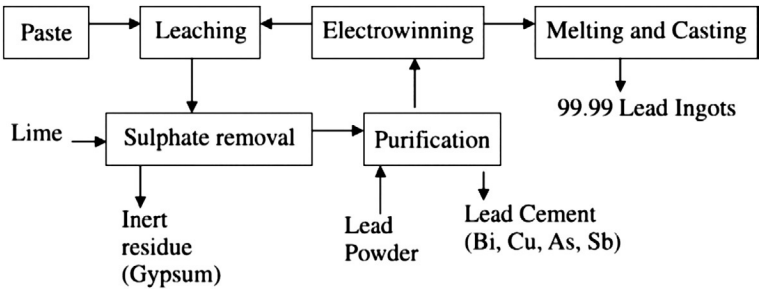


FIG. 16

Flow diagram of the PLACID process.

Reproduced from Zhang W. Yang J. Wu XW, Hu Y, Yu W, Wang J, Dong J, Li M, Liang Sh, Hu J, Kumar RV. A critical review on secondary lead recycling technology and its prospect. Renew Sustain Energy Rev 2016;61:108–122.

The average purity of lead obtained from the PLACID process was measured as 99.995 wt.%, and the recovery ratio of Pb was more than 99.5 wt.%. However, the energy consumption was nearly 1300 kWh per ton of lead output [24].

Life cycle assessments for LABs

From the results of studies conducted by the International Lead Association (ILA) on life cycle assessments (LCA) and life cycle inventories (LCI) [23] there were the following conclusions:
Lead metal production LCI:

- Mining and smelting have the greatest environmental impacts on lead production.
- The main contributors to mining and concentration are fuel combustion and power production.
- The study represented 80% of production technology but only 32% of ILA members.

Lead-based batteries LCA:

- Lead production (from ores or recycled scrap) is the dominant contributor to environmental impacts associated with the production of lead-based batteries.
- The high recycling rates associated with lead–acid batteries dramatically reduce any environmental impacts.
- In terms of global warming potential, the environmental advantage of improved and advanced technology lead-based batteries during the use phase far outweighs the impacts of their production.

Safety issues of lead–acid batteries

Thermal safety problems

The optimal operating temperature range of lead–acid batteries is 20–40°C.

Lower temperatures limit both its charging capability and general electrical performance. Furthermore, in dependence of the SOC and the battery type (electrolyte amount), a freezing of the electrolyte is possible. For SLI batteries typical values are:

- 100% SOC about −70°C
- 50% SOC about −20°C
- 0% SOC about 0°C

Frozen batteries can burst, and acid will leak out once it is thawed. Charging of frozen batteries can also cause explosions [25].

The upper-temperature limit for continuous operation measured inside the battery is 60°C for flooded and 55°C for AGM batteries. Higher temperatures increase strongly self-discharge, wear-out, and water consumption which can lead over a longer time period to the destruction of the battery. But a short time increase of the temperature beyond the maximum operating temperature does not bring safety problems.

A battery temperature increase occurs when the heat production is greater than the heat dissipation into the environment and the heat storage capacity of the battery [26].

$$T = f\left(\textit{heat release, heat exchange, heat adsorption}\right)$$

Heat release

Heat release occurs via Joule loss heat (Q_{Joule}), and the thermodynamically determined reversible heat effect ($Q_{\text{rev}} = T\,\Delta S$; 11.7 kJ for LAB).

$$Q_{\text{Joule}} = I^2\,R$$

R: all ohmic resistances in the battery and electrochemical polarizations
I = Charge and discharge current.

Heat exchange

$$Q_{\text{exchange}} = q_{\text{exchange}} \times t$$

q_{exchange}: heat exchange rate (kJ/s).

$$q_{\text{exchange}} = q_{\text{heat radiation}} + q_{\text{heat convection}} + q_{\text{heat conduction}}$$

Heat adsorption

$$Q_{\text{ad}} = Q_{\text{cp}} \times \Delta T$$

Q_{cp}: Heat capacity $Q_{\text{cp}} = C_{\text{p}} \times m$ of acid, active masses, grid, container, etc.

The temperature change of the battery (ΔT) is therefore determined by.

$$\Delta T_{batt} = \frac{I^2 R + T\Delta S - Q_{exch}}{m\,C_p}$$

Thus, the load profile (I) for a given battery type (R, C_{p}) and heat transfer (Q_{exch}) are the decisive factors.

A further, atypically temperature increase can occur if the operating window is left upwards caused by defects that have arisen as a result of battery aging and faulty manufacture, and if the battery is heated externally (Fig. 17).

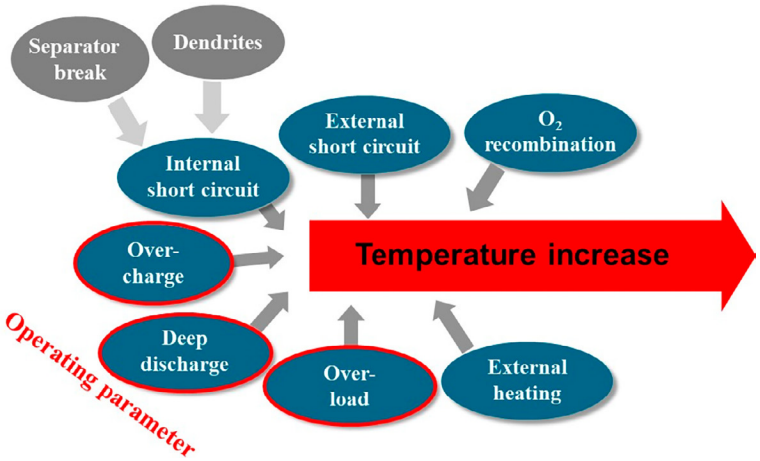


FIG. 17

Additional untypically reasons for temperature increase of batteries.

The temperature increase of the cell in case of a short circuit could reach theoretically (adiabatic conditions, complete reaction) based on a heat capacity (C) of 1.15 kJ/kg/K and a specific energy (E) of 30 Wh/kg (108 kJ/kg) for a stationary battery, 110 K (E/C). The practical values, however, are lower, caused by temperature exchange and not complete discharge reactions.

The trigger for a short circuit could be externally a low resistance connection of both terminals or internal dendrite formation or separator break as a consequence of material fatigue or strong mechanical pressure. Furthermore, short circuits occur after aging by mossing, short circuits across the mud chamber, growth in the separators, or the formation of a conductive acid film on the surface of the battery cover due to gassing with a reduction of the insulation resistance.

High ambient temperatures are a source for an increase of the internal battery temperature as well, especially in warm countries and for SLI batteries in the hot engine car compartment. Furthermore, a stronger heating is observed of the middle cells compared to the outer cells of the battery, especially at AGM batteries caused by their lower heat capacity and oxygen recombination. Therefore, it is important to pay attention to the temperature differences of cells in particular at AGM batteries, where the differences between the cells should not exceed 5 K [25].

Charge/overcharge (see Fig. 18), overdischarge [27], and high currents could lead to a temperature increase as well.

The Joule heat (Q_{joule}) plays an important role in the heat release. The Joule heat development (dQ_{joule}/dt) is given beside the current by the difference between the terminal voltage (U_{tm}) and the equilibrium voltage (U_{o}).

$$dQ_{\text{joule}} / dt = i\left(U_{\text{tm}} - U_{\text{o}}\right)$$

The equilibrium voltage of the LAB battery ($U_{\text{o}} = 2.12$ V), however, is changing during overcharge for flooded batteries to 1.23 V and for VRLABs to 0 V—see later. This leads to a strong increase of the Joule heat of VRLABs.

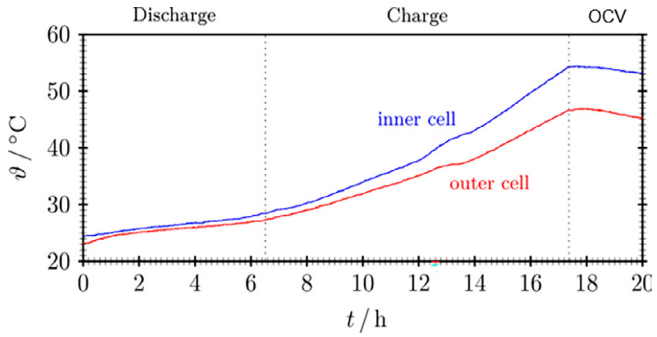


FIG. 18

Temperature of an inner and outer cell of a Hoppecke 80 V 5 PzS 700 battery during discharge, charge/overcharge, and following OCV period.

The equilibrium potentials/voltage (φ_o, U_o) in LAB cells with an acid density of 1.28 g/cm^{-3} for the positive and negative electrode and the cell are given in Eqs. (16)–(18).

Positive electrode

$$PbSO_4 + 2H_2O \Rightarrow PbO_2 + 4H^+ + SO_4^{2-} + 2e^- \quad \varphi_o = 1.73\,V \tag{16}$$

Negative electrode

$$PbSO_4 + 2e^- \Rightarrow Pb + SO_4^{2-} \quad \varphi_o = -0.39\,V \tag{17}$$

Cell

$$2PbSO_4 + 2H_2O \Rightarrow PbO_2 + Pb + 2H_2SO_4 \quad U_o = 2.12\,V \tag{18}$$

For the overcharge of flooded LAB batteries is valid:

Positive electrode

$$2H_2O \Rightarrow O_2 + 4H^+ + 4e^- \quad \varphi_0 = 1.23\,V \tag{19}$$

Negative electrode

$$2H^+ + 2e^- \Rightarrow H_2 \quad \varphi_o = 0\,V \tag{20}$$

Cell

$$2H_2O \Rightarrow O_2 + 4H^+ + 4e^- \quad U_o = 1.23\,V \tag{21}$$

However, the overcharge of VRLA batteries proceeds as follows:

Positive electrode

$$2H_2O \Rightarrow O_2 + 4H^+ + 4e^- \quad \varphi_o = 1.23\,V \tag{22}$$

Negative electrode

Caused by its more positive potential the O_2 reduction is preferred over the hydrogen development (Eq. 20).

$$O_2 + 4H^+ + 4e^- \Rightarrow 2H_2O \quad \varphi_o = 1.23\,V \tag{23}$$

Besides the direct electrochemical reaction path it is also possible

$$\begin{aligned}
&O_2 + 2Pb \Rightarrow 2PbO && \text{chemically}\\
&2PbO + 2H_2SO_4 \Rightarrow 2PbSO_4 + 2H_2O && \text{chemically}\\
&2PbSO_4 + 4e^- \Rightarrow 2Pb + 2SO_4^{2-} && \text{electrochemically}\\
&\mathbf{O_2 + 4H^+ + 4e^- \Rightarrow 2H_2O} && \textbf{brutto reaction}
\end{aligned} \tag{24}$$

Cell

In the cell no brutto cell reaction takes place in the overcharge phase. The oxygen generated at the positive electrode is reduced at the negative electrode (oxygen cycle)

$$2H_2O \Rightarrow O_2 \Rightarrow 2H_2O \; U_o = 0\,\text{V} \tag{25}$$

In these reaction considerations, the grid corrosion is neglected, because their influence on the electrode potential is low.

In the following observations on heat generation during charging and overcharging, the reversible heat effect ($T\Delta S$) was neglected due to the dominance of Joule heat. While the equilibrium cell voltage of flooded and VRLA batteries amounted to 2.12 V, the equilibrium voltage for the overcharge reaction (water decomposition) of flooded batteries is 1.23 V, and of VRLA batteries (oxygen cycle: oxygen generation at the positive electrode, oxygen reduction at the negative electrode) is 0 V.

For example, the Joule heat development when charging with Uch = 2.5 V and also overcharging with Uovch = 2.5 V would amount for various LAB types to:

Charge reaction of flooded and VRLA batteries:

$$Q_{\text{Joule}} = i\left(U_{\text{ch}} - U_o\right)t = i \times \left(2.5\text{V} - 2.1\text{V}\right)t = i \times \mathbf{0.4V} \times t \tag{26}$$

Overcharge reaction of flooded batteries

$$Q_{\text{Joule}} = i\left(U_{\text{ovch}} - U_o\right)t = i \times \left(2.5\text{V} - 1.23\text{V}\right)t = i \times \mathbf{1.27V} \times t \tag{27}$$

Overcharge reaction of VRLA batteries

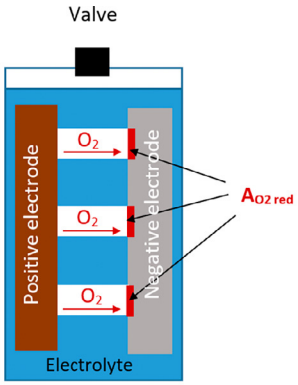
$$Q_{\text{Joule}} = i\left(U_{\text{ovch}} - U_o\right)t = i \times \left(2.5\text{V} - 0\text{V}\right)t = i \times \mathbf{2.5V} \times t \tag{28}$$

That means the largest voltage difference $U_{\text{tm}} - U_o = 2.5\,\text{V}$ and therefore the largest Joule heat are observed in case of overcharge of VRLA batteries (Eq. 28). This is valid if during overcharge all oxygen generated with $i_{O2\,\text{gen}}$ at the positive electrode (Eq. 22) will be reduced with the current $i_{O2\,\text{red}}$ at the negative electrode (Eq. 23), i.e., $i_{O2\,\text{gen}} = |I_{O2\,\text{red}}|$. In this case the cell voltage will be 0 V (Eq. 25).

The $i_{O2\,\text{gen}}$ is constant at constant current charge and given by the overcharge current (i_{ovch}), i.e., $i_{\text{ovch}} = i_{O2\,\text{gen}}$ the O_2 reduction current is determined by the O_2 diffusion through gas pores in the immobilized electrolyte, and the O_2 reduction area ($A_{O2\,\text{red}}$) (Fig. 19).

Provided of gas pores in the immobilized electrolyte this surface $A_{O2\,\text{red}}$ is given by the area of the partially gas-filled negative active mass pores at the electrolyte side. The oxygen reduction current, however, is given besides the O_2 reduction area also by the oxygen diffusion length, i.e., by the thickness of the electrolyte film on the Pb surface (wetting film thickness). The wetting film thickness as well as the O_2 reduction area depend on the radius of the gas-filled pores of the negative active mass pores [28]. With decreasing electrolyte saturation of the VRLA cell even smaller pores are available for the O_2 reduction, i.e., will be filled with gaseous oxygen. Smaller pores hold the electrolyte longer due to capillary forces.

Both parameters are shown in Fig. 20 in dependence on the electrolyte saturation level.



$$2H_2O \Rightarrow 4H^+ + 4e^- + O_2 \Rightarrow O_2 \Rightarrow O_2 + 4H^+ + 4e^- \Rightarrow 2H_2O$$

FIG. 19

Working principle of a VRLA battery in the overcharge phase; $A_{O2\,red}$: oxygen reduction area (for clarity no separator is shown).

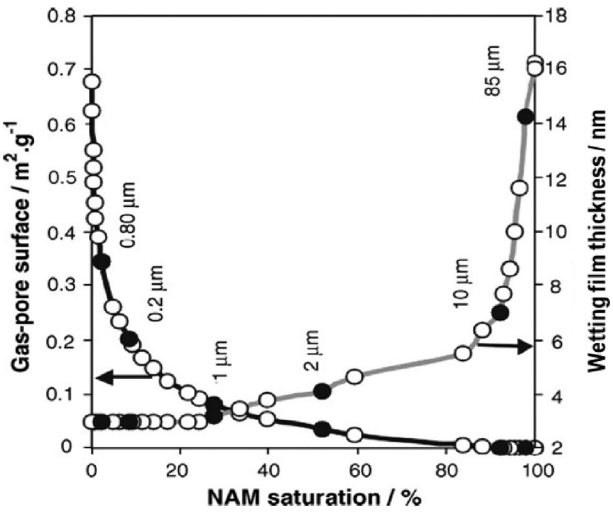


FIG. 20

Dependence of the area of the gas-filled negative active mass pores ≡ $A_{O2\,red}$ *(black curve)* and wetting film thickness *(gray curve)* on the electrolyte saturation. The *black dots* correspond to the listed pore radii.

Reproduced from Kirchev A, Pavlov D. Influence of temperature and electrolyte saturation on rate and efficiency of oxygen cycle in VRLAB. J Power Sources 2006;162:864–869.

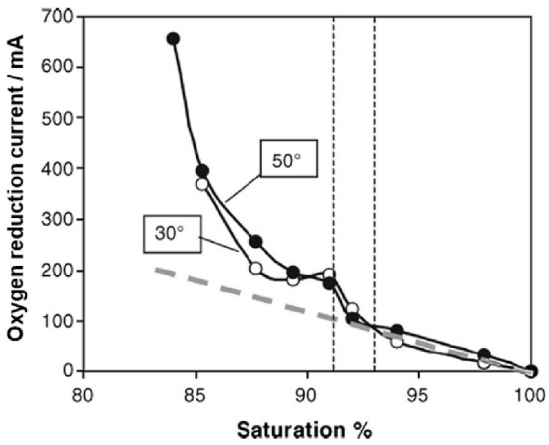


FIG. 21

Dependence on the oxygen reduction current of the electrolyte saturation at $\phi^{-} = -1.2$ V (vs Hg/Hg_2SO_4), at 30°C and 50°C.

Reproduced from Kirchev A, Pavlov D. Influence of temperature and electrolyte saturation on rate and efficiency of oxygen cycle in VRLAB. J Power Sources 2006;162:864–869.

Fig. 20 shows, that at a saturation level of <80% the area of the gas-filled pores increases exponentially and therefore also exponentially the oxygen reduction current (Fig. 21).

The influence of the saturation effect on the H_2 generation, O_2 recombination, and parasitic reactions is shown in Fig. 22. It is to be mentioned that the saturation is also a function of the cycle lifetime because initially a larger gas outlet is observed, caused by no/low oxygen recombination and leading to a saturation decrease (see also later).

An increasing O_2 reduction on the lead surface causes the negative plate potential to shift in the positive direction, which increases the positive plate potential in the constant voltage (CV) charge and accelerates the oxygen evolution further.

The relevant battery temperature increase depends not only on the oxygen recombination rate which determines the Joule heat development but also on the ambient temperature, the battery thermal capacity (depending on the specific electrolyte content—mL Ah^{-1}) and the heat dissipation (Q_{disp}) (cooling).

If $Q_{joule} \approx |Q_{disp}|$ the temperature will be constant at a relative high level (Fig. 23A). If the generated heat, however, is larger than the dissipated heat, $Q_{joule} > |Q_{disp}|$ then the temperature will increase continuously and the system will enter a thermal runaway situation (Fig. 23B).

The thermal runaway [5,10] of LAB batteries is not comparable in its consequence with Li-ion batteries. The maximum temperature in a thermal runaway event of LABs is <100°C and does not lead to the fire and explosions known for Lithium-ion batteries [29].

This thermal runaway situation is promoted at aged VRLA batteries because during the cycle life, some oxygen and also hydrogen have left at unfavorable oxygen reduction conditions in the cell. A gas outlet can be observed also under dynamic charge conditions with high current pulses, where the gas pressure inside the cell reaches the valve opening pressure before the oxygen is reduced.

This leads to lower electrolyte saturation of the AGM/gel cell, i.e., larger O_2 reduction areas and thinner wetting film thickness (see Fig. 20) and so to higher oxygen reduction current. Therefore the risk of thermal runaway is growing with the age of VRLA batteries.

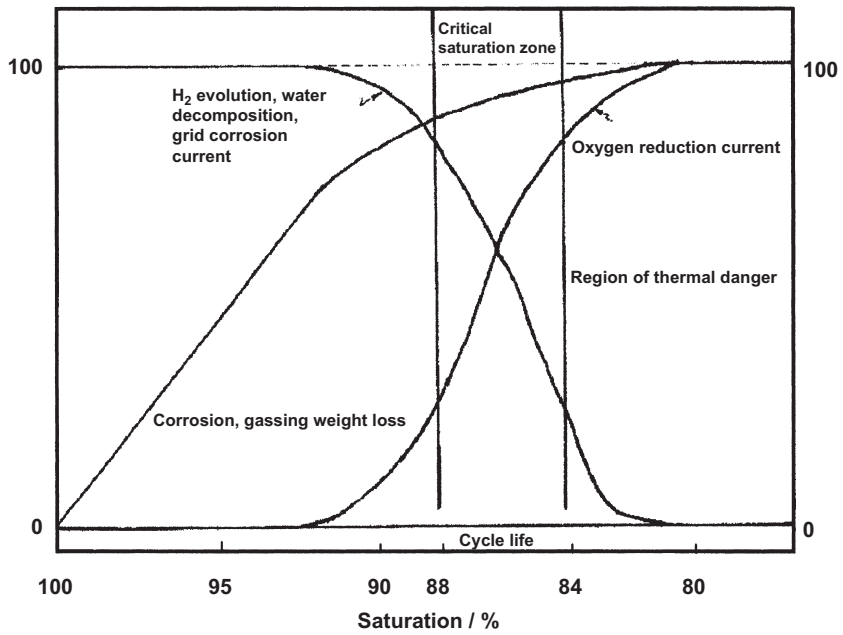


FIG. 22

Relative distribution of the overcharge current a function of the AGM saturation.

Reproduced from Nelson RF. Charging techniques for VRLA batteries. In: Rand DAJ, Moseley PT, Garche J, Parker CD, editors. Valve regulated lead-acid batteries. Elsevier; 2004 [chapter 9].

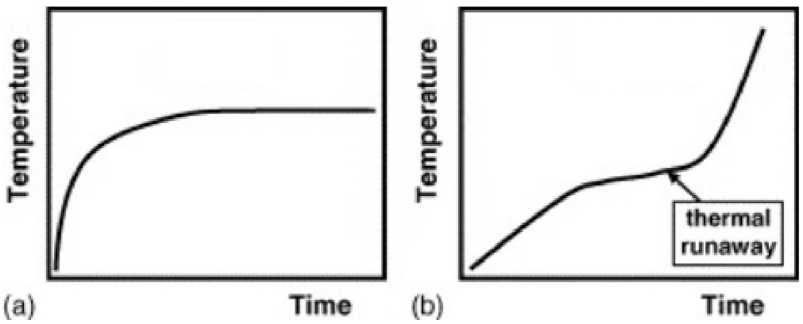


FIG. 23

Temperature changes in VRLA batteries during overcharge, (A) normal temperature behavior, (B) thermal runaway behavior.

Reproduced from Pavlov D, Monahov B, Kirchev A, Valkovska D. Thermal runaway in VRLAB—Phenomena, reaction mechanisms and monitoring. J Power Sources 2006; 158:689–704.

A thermal runaway can be observed also locally. One reason for that is an uneven filling of the AGM, where local differences in the degree of filling could be not compensated by the capillary properties of the AGM. Another reason is local short circuits lead to local heating and a local decrease in the degree of filling, thus causing an increased local transport of oxygen, a higher recombination, and further local heating, which is triggering a thermal runaway.

Because the risk of a stronger temperature increase and finally a thermal runaway is given mainly by the size of the O_2 reduction areas and the wetting film thickness of the negative electrode in VRLABs. But such strong temperature increase can also occur at flooded LAB batteries, if the electrolyte level is decreased below the plate upper edge. The reason for this is electrolyte water decomposition during overcharge (Eqs. 19 and 20) and the nearly total release of the gases via the plug without recombination (see Fig. 24).

This electrolyte level decreasing below the plate's upper edge creates also O_2 reduction areas on the negative electrode. That is, there are two reactions at the negative electrode of partially dried-out flooded cells, the oxygen reduction (Eq. 23) at the upper part with no electrolyte between the plates and the hydrogen evolution in the lower part with electrolyte between the plates (Eq. 19)—see Fig. 24.

Normally, flooded batteries with an electrolyte level slightly under the plate's upper edge, do not tend to a thermal runaway; their higher specific electrolyte volume (mL Ah^{-1}) leads to a higher heat capacity, larger heat transport to the container wall, and larger evaporative cooling.

But with further electrolyte level decrease during the lifetime more and more O_2 reduction areas ($A_{O2\,red}$) on the negative electrode are formed. This is in a first approximation similar to a reduction of the electrolyte saturation (see Fig. 20) and so also a thermal runaway situation could be reached.

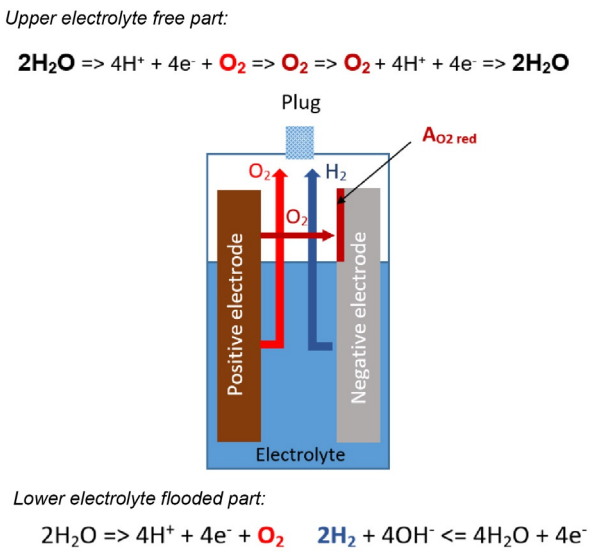


FIG. 24

Working principle of a flooded, partially dried-out LAB battery in the overcharge phase (for clarity no separator is shown).

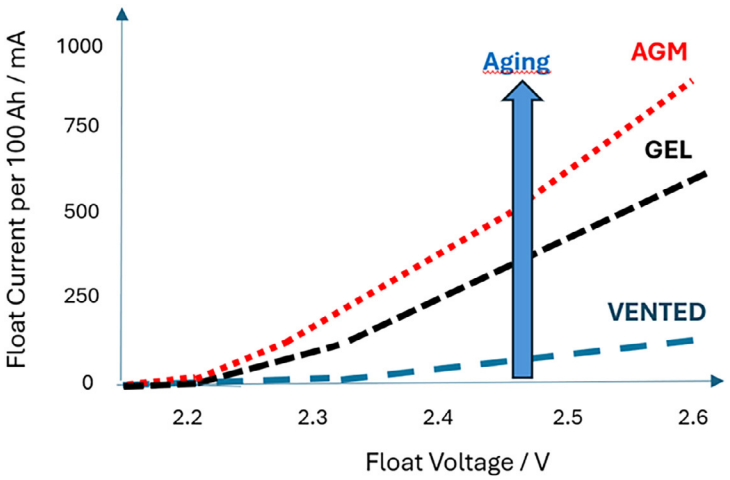


FIG. 25

Float current vs. float voltage for different LAB battery types at 25°C.

This is especially observed at larger batteries (truck batteries, industrial batteries), where the heat dissipation in relation to the heat generation is reduced in the presence of impurities that lower the oxygen and notably also the hydrogen overvoltage, hereby promoting water loss. Such impurities are often Sb-containing species in batteries with PbSb grids. In the case of older forklift PbSb batteries thermal runaway is well known for cases when the U_{absorb} stage in the IUI charging regime is not reached due to the Sb-induced reduction of the charging voltage. In this case, the charger continues the charging at full current because during the first I_{const} (bulk) stage it did not reach the U_{absorb} voltage and was impeded to automatically skip to the next (absorb) stage.

The influence of the oxygen reduction area and the thickness of the electrolyte film of the negative electrode on the float current, depends on the float voltage is shown in Fig. 25, whereby $A_{O_2\,red}^{AGM} > A_{O_2\,red}^{Gel} > A_{O_2\,red}^{vent}$ and all $A_{O_2\,red}^{PbA}$ can be increased with growing age.

A thermal runaway situation does not occur at normal operating conditions, i.e., if temperature and charge voltage limits are not exceeded, the electrolyte level is above the upper plate edge (flooded) and the electrolyte saturation level is not lower than 85% (VRLAB).

The float voltage has a strong influence on the oxygen and hydrogen evolution. When the voltage is <2.4 V the oxygen reduction at the negative plate depolarizes the negative electrode, and the hydrogen evolution is very slow. The potential of the negative electrode is mainly determined by the oxygen recombination rate rather than by the applied voltage. Therefore, the applied voltage mainly affects the oxygen evolution. Its Tafel slope is about 120 mV per decade. The oxygen evolution current increases exponentially with its overpotential. A high voltage applied to the battery can obviously accelerate water loss, oxygen cycle current, temperature rise, and even lead to thermal runaway [5,10].

The following problems are discussed which are related to the temperature increase of LAB batteries. As already mentioned, the maximum operating temperature of lead–acid batteries is about 60°C (55°C AGM).

If the battery temperature is increased to >70°C the jars will soften. A measure for the critical temperature which should not be exceeded is the *heat deflection temperature* (HDT) which is related to the melting temperature of the container.

The heat deflection temperature for different battery container materials is listed as follows:

PP (not filled)	HDT = 70 … 95°C
PP (filled)	HDT = 90 … 120°C
ABS (not filled)	HDT = 79°C

General requirements for the SLI battery container are given in the standard *EN50342-5 Lead–acid starter batteries—Part 5: Properties of battery housings and handles.* Exceeding the HDT leads to a bulging of the container.

At temperatures of about 120°C and when electrolyte evaporation is observed the battery electrolyte dries out, leaks, and there is a risk of acid mist. As long as there is still electrolyte liquid, the temperature does not rise above 120°C, because the H_2O evaporation prevents further temperature rise. With the complete evaporation of the water the internal resistance rises strongly and reduces the current which prevents a further temperature increase. The remaining highly concentrated H_2SO_4, however, oxidizes the separator.

At very high current flow an overheating of the terminals and straps is observed leading to softening of the battery cover around the terminals and intercell connections.

In the case of underdimensioned top lead components can even meltdown, i.e., in a worst case with unusually high currents, jar bulging, jar meltdown, and in extreme cases, cell explosions and fires can happen.

At temperatures around 160°C the melting point temperature (T_{mp}) of the container materials will be reached.

PP	T_{mp} = 160 … 175°C
ABS	T_{mp} = 130 … 160°C

At the melting point, a rupture of the container takes place, and at temperatures >300°C it can cause a fire, which is not related to hydrogen. The reason for this non-hydrogen fire is the decomposition of container materials under the formation of combustible gases. The decomposition temperature (T_{dc}) is as follows:

PP	$T_{dc} \geq 380$°C
ABS	$T_{dc} \geq 300$°C

In general, the requirements for burn resistances especially for these container materials used in occupant compartments of passenger cars are standardized, see, e.g., US FMVSS 302; EG 661/2009, and internal standards of car companies.

Kinetically safety problems

Oxyhydrogen gas is the reason for the explosions of LAB batteries leading to blast waves and material ejections (mostly casing parts and H_2SO_4 electrolyte). The source for the oxyhydrogen gas is hydrogen gas formation at the end of charge, overcharge, and also self-discharge.

At room temperature flammability limits of H_2-mixtures with air are 4.0%–75.0% H_2 and the explosion limits are 18.3%–59% H_2, whereby the ignition energy is with only 0.02 mJ very low; this energy is practically overall available.

If, however, a spark-ignited fire or explosion does not take place, the hazard can alternatively be caused by the self-ignition of the 2:1 H_2:O_2 mixture at about 570°C with a reaction enthalpy of 286 kJ/mol.

The hydrogen development rate depends on the charge/overcharge voltage and temperature (Fig. 26).

Furthermore, the alloy (see Fig. 27), additives, and impurities (see, e.g., IEC 62877-1,-2 to allow contaminations of acid/water) take influence on the hydrogen overvoltage and therefore on the hydrogen generation rate.

As already described earlier, the ignition energy for an oxyhydrogen gas explosion is with 0.02 mJ very low and therefore many ignition sources are normally available in the battery surroundings, as sparks and arcs formed by electrical equipment, switches or fuses, soiling poles/connections, metallic objects on the battery poles.

Usually the ignition source is located outside of the battery and therefore the oxyhydrogen gas formed from leaked hydrogen and air oxygen will be ignited also outside of the battery, provided that the hydrogen is not yet diluted and its concentration at the ignition source lies in the fire/explosion range within the surrounding air. The distance from the battery until the place, where the hydrogen concentration falls below the fire/explosion limit is the critical distance (d_{cr})—see Fig. 28.

It is to take care, that at least this critical distance (d_{cr}) is present between battery and potential ignition sources (electronic equipment, hot engine, etc.).

The distance between battery and ignition source and necessary ventilation of installation rooms are described in the standards EN50272/IEC 62485.

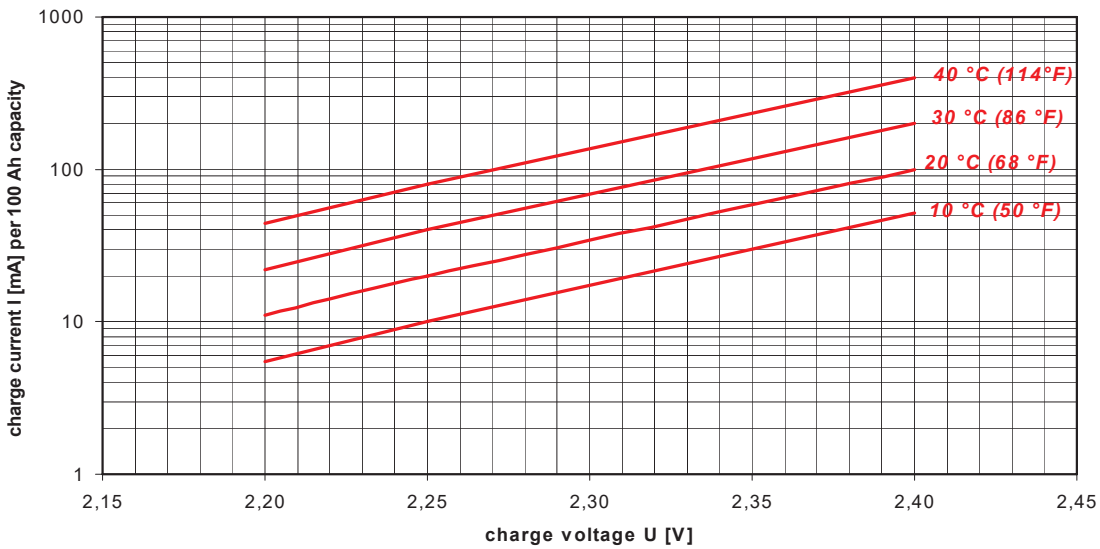


FIG. 26

Impact of temperature on charging current & voltage, for new LAB batteries with Pb-Sb-Se grids.

Reproduced from O'Donnell C, Schiemann M. Hydrogen gas management for flooded lead acid batteries. In: Proceedings of the Battcon Conference in Orlando, FL, USA; 2008.

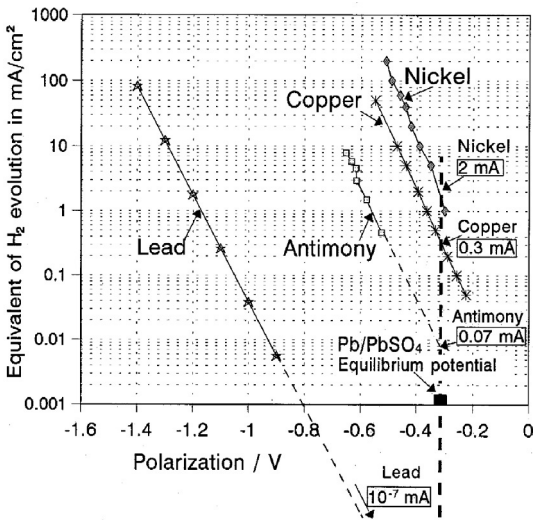


FIG. 27

Hydrogen evolution rate on different metals in dependents on polarization.

Reproduced from Berndt D. Jones DEW. Valve-regulated batteries in standby application, In: TELESCON 2000. Third International Telecommunications Energy Special Conference, IEEE; 2000. p. 89–96. Courtesy of Jones WEM. Philadelphia Scientific International.

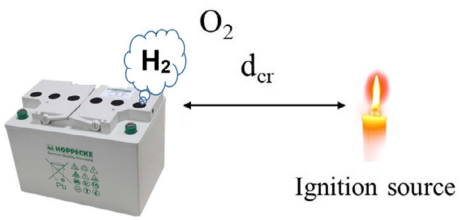


FIG. 28

Schematic representation of the critical distance (d_{cr}) between battery and ignition sources.

To increase the distance of an ignition source to the battery the car jump start rule is: Positive cable clamp to the positive terminal on the battery and negative cable clamp to a bare metal part of the engine block. The critical distance could be reduced if a well-positioned battery gas outlet hose is available.

If an outer ignition source sets off a battery outside fire/explosion it is possible that a flash-back into the battery occurs via pores at lid/box heat sealing areas, not proper sealed plugs and valves as well as areas of too thin walls at lid or box (EDC proof with 10 kV necessary). At the most lead–acid batteries, however, the gas outlet opening is protected by the installation of a flame arrestor. Thus, the ignition of the battery by flames and sparks from the outside is been largely prevented.

There are also situations where fire/explosions could occur inside the battery if a flammable/explosive oxyhydrogen gas and sparks or arcs exist inside the battery container. Sparks or arcs could be formed inside the battery if positive and negative plates come into contact, e.g., after corrosion-induced

growth of the positive grid or tear off of Pb connectors of the negative electrode block (lugs, busbar, terminal) resulting from manufacturing errors, vibrations, corrosion (if the connectors are not covered with electrolyte—especially the lugs).

As internal ignition source can also act an internal electrostatic charge. Such an electrostatic charge up to 30 kV could be formed by friction if the battery container lid is wiped with a dry cloth. The electrostatic charge can be partially transferred on a single electrolyte drop sticking inside of the lid as a hidden conductor preferably at dry-charged batteries. The LAB cell can act as grounding for a discharge of the electrostatic charged source. If the charged drop is falling into the acid batch the oxyhydrogen gas inside could be ignited, leading to an explosion.

Oxyhydrogen explosions could be prevented via

- reducing the hydrogen generation rate by
  - not too high ambient temperatures; using batteries with normal electrolyte level (vented) and saturation (AGM)
  - not too high charge voltages (<2.45 V @ RT); it is to prove whether alternator voltage limitations are possible
  - filling up the electrolyte with distilled water without pollutants
- preventing ignitions by
  - good ventilation of battery rooms
  - taking care of the critical distance between the battery and the possible ignition source
  - prevention of electrostatic charges during handling of batteries (production, installation) [25]

At normal operating conditions, *oxyhydrogen gas explosions* are not to be feared!

Electrical safety problems

The effect of current passing through the human body depends on the current intensity, voltage, contact area, contact pressure, and body resistance. Human body resistance varies with contact area, contact pressure, thickness of skin, and presence of moisture. The perception threshold of current is 0.2–1 mA, 10–15 mA causes muscle reaction, 15–25 mA can produce diaphragm spasm, and 60–100 mA a ventricular fibrillation. Currents through the body of about 1 A cause a heart stop. Such high currents hardly occur if the battery voltage is less than 60 V, but some EV, and especially BESS batteries could have a much higher voltage. So, the safety requirements are much stronger—see the standards EN50272_2 Chapters 5 and 10 as well as IEC 62485-2 Chapters 4 and 9.

Chemical safety problems

As already mentioned chemical safety problems are dealt with in the earlier sections *Issues related to lead and lead compounds, sulfuric acid electrolyte, and toxic gases* due to their similarities with environmental problems.

Acknowledgments

This chapter was written with the very strong support of Dr. Manfred Gelbke (MG consulting, Bad Staffelstein—Germany).

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